

SMART INDIA HACKATHON 2026



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HACKATHON
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TITLE PAGE

- **Problem Statement ID** – SIH26104
- **Problem Statement Title** - AI-Powered Real-Time Detection and Prevention of Voice Cloning Impersonation Attacks
- **Theme** - Miscellaneous
- **PS Category** - Software
- **Team ID** - 140037
- **Team Name (Registered on portal)** - Hackspire2..0

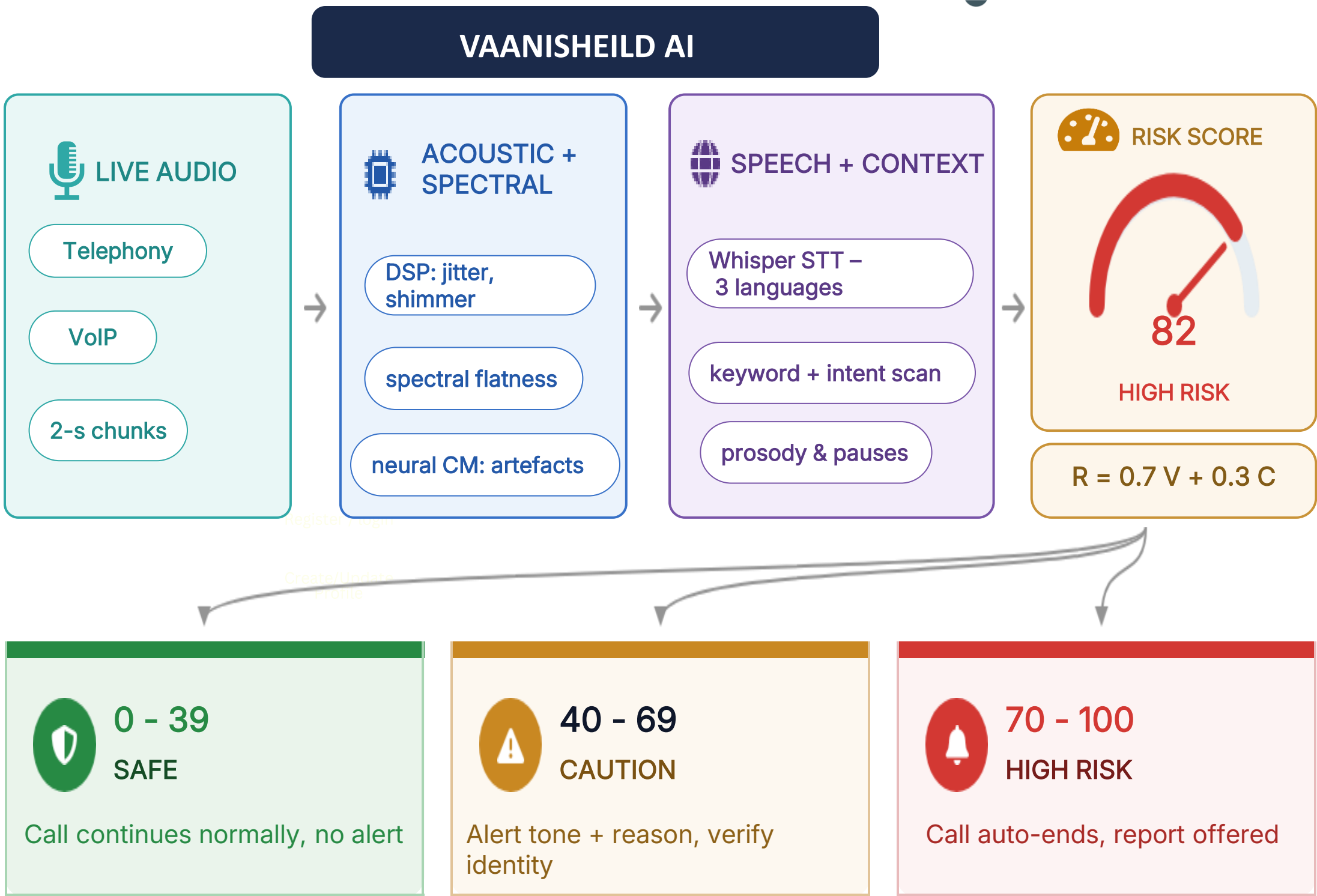


PROBLEM OVERVIEW & EXPECTED SOLUTION

- **Problem:** AI-cloned voices bypass traditional call verification, enabling real-time fraud via impersonation calls.
- **Solution:** Real-time voice-authenticity engine that scores impersonation risk live and alerts before sensitive actions occur.

PROPOSED SOLUTION

- **Background Call Guardian :** Live floating risk badge on the call screen - Truecaller-style, no manual upload.
- **Graduated Live Response :** Green (Safe) continues , Yellow beeps + warns , Red (High-Risk) auto-ends the call.
- **Real-Time Voice Deepfake Detection:** DSP + AI models analyze 2s audio chunks for cloned/spoofed voice signatures.
- **Multilingual & Accent-Aware:** Hindi, English, Marathi support for Indian users and regional accents.
- **Context-Aware Risk Fusion:** Combines voice authenticity with NLP keyword risk (OTP, transfer, urgent, digital arrest).
- **Explainable & Privacy-First:** “Why Flagged?” reasons on a dashboard; minimal retention, edge/on-device option.



TECH STACK

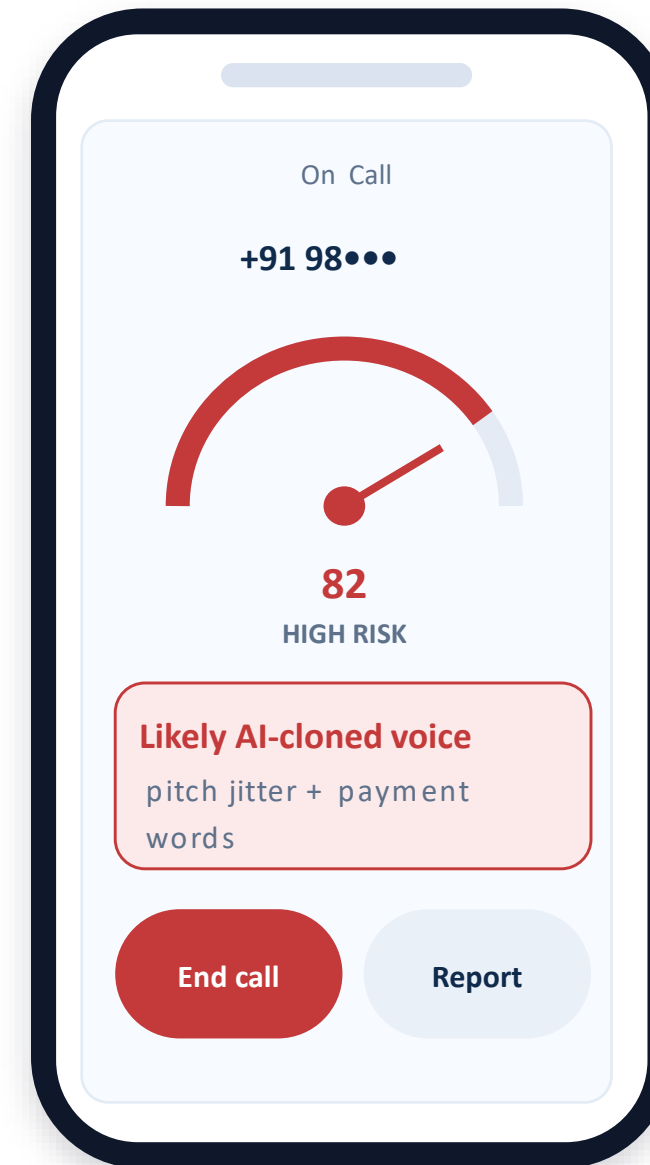
- › **Mobile App:** Java / Kotlin
- › **Backend:** FastAPI, Python, Uvicorn
- › **Voice / Speech AI:** Librosa (DSP), Whisper (multilingual STT), AASIST/RawNet2
- › **Data & Storage:** PostgreSQL, Redis, SQLite
- › **Infra:** Docker, Render, Vercel, Edge/TFLite
- › **Security:** AES-256, JWT/OAuth2, DPDP-aligned, audit logs

COMPOSITE RISK FORMULA

$$\text{Risk} = 0.7 \times \text{VoiceScore} + 0.3 \times \text{ContextScore}$$

VoiceScore = f(pitch jitter, shimmer, spectral flatness, HF-energy ratio) per 2s chunk

ContextScore = min(100, 25 * matched risk-keywords in live transcript)



WHAT YOU GET

- **Warning before the ask**
risk moves past 40 mid-sentence
- ✓ **Explainable, never a black box**
reason codes on every score
- ✗ **No Audio Retention**
features only, discarded at call end
- ⚡ **Runs on the edge**
~2 s re-scoring, no cloud round-trip

 **Multilingual**
• Hindi • English
• Marathi

 **Explainable**
“Why Flagged?”

 **Privacy-first edge**
processing

 **B2B API + consumer**
app

 **Live Prototype Link :** [click here](#)

 **Youtube Link :** [click here](#)

 FEASIBILITY
→ **Operational Feasibility**

- Easily integrates with existing **bank & telecom** systems.
- Works in real-time without disrupting existing workflows.**

→ **Economic Feasibility**

- Open-source + CPU-based system keeps costs low.
- Can scale without expensive GPU infrastructure.

→ **User Feasibility**

- Automatic detection with minimal user effort.
- Simple, multilingual alerts** help non-technical users.

→ **Technical Feasibility**

- With **Digital Signal Processor** + Whisper/Librosa detection engine
- 3 access paths:** Default Dialer , Telecom Integration , VoIP/In-App
- API-first, CPU-based & cloud/edge deployable

 VIABILITY
→ **Technical & Operational Viability**

- Detection system can **adapt to new voice-cloning techniques.**
- Flexible deployment across **Device, Telecom & VoIP.**

→ **Economic Viability**

- Multiple revenue streams:** Banks, telecom operators & consumer app.
- Low operating cost with **scalable deployment.**

→ **Social & Legal Viability**

- Privacy & consent-first** approach.
- Helps protect **elderly and vulnerable users** from voice scams.

→ **Scalability**

- One core engine can support **multiple platforms.**
- Can expand from **pilot** → **banks** → **telecom** → **large-scale deployment.**

HOW WE ACCESS CALL AUDIO : PLATFORM FEASIBILITY**Path A: Default
Dialer App**

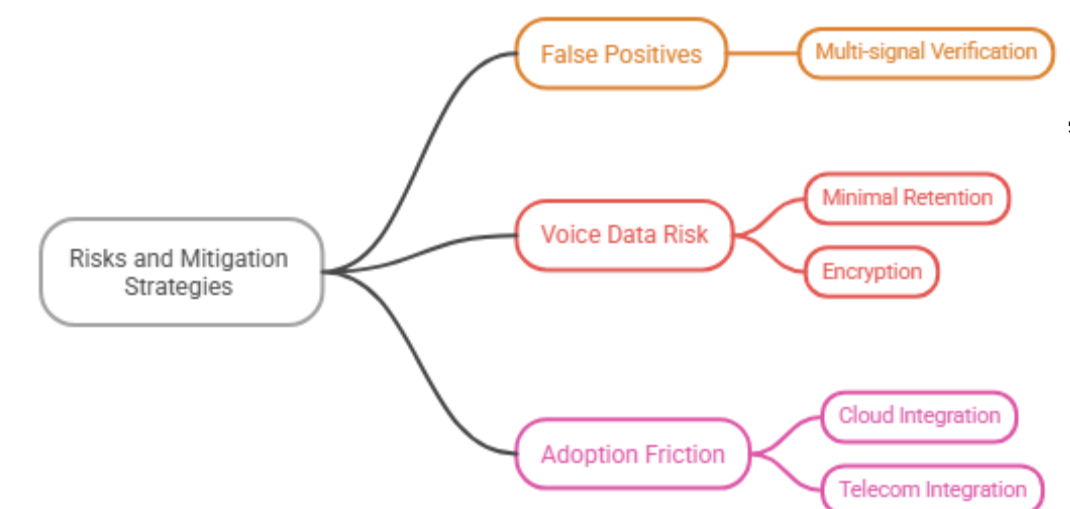
Android InCallService gives real call-audio access when VaaniShield is set as the phone app.

**Path B: Telecom-
Side Integration**

Carriers integrate our detection API at the network level - most scalable, no install needed.

**Path C: VoIP / In-
App Calling**

Direct audio access with no OS permission battle - used for this hackathon's live demo.

**POTENTIAL
RISK AND
MITIGATION
STRAGIES**


ImpactsReduced Financial
FraudStronger Cyber
ResilienceSafer Digital
PaymentsProtection from
Vulnerable Users**IMPACT****REDUCED FINANCIAL FRAUD**

Prevents or interrupts voice-based scams before financial loss occurs.

STRONGER CYBER RESILIENCE

Adds a proactive defense layer against AI-generated voice and social-engineering attacks.

SAFER DIGITAL PAYMENTS

Builds greater confidence in phone-based banking and digital transactions.

PROTECTION FROM VULNERABLE USERS

Provides an additional safety layer for senior citizens and less digitally experienced users.

From Detection to Prevention

Shifts fraud protection from “*detect after the incident*” to “*intervene before the damage.*”

BenefitsReal-Time
Protection

Explainable Alerts

Multilingual Security

Privacy-First

Fraud Prevention

BENIFITS**REAL TIME PROTECTION**

Scam calls can be flagged within seconds, before the victim transfers money.

EXPLAINABLE ALERTS

Users understand *why* a call is risky instead of receiving only a generic risk score.

MULTILINGUAL SECURITY

Supports Hindi, Marathi and English, making protection accessible to a wider population.

PRIVACY - FIRST

Designed around minimal audio retention and on-device/edge processing where feasible.

Fraud Prevention

Helps users pause, verify and avoid potentially fraudulent transactions.

AI / DEEFAKE RESEARCH

AASIST – Audio Anti-Spoofing

Audio deepfake/spoof detection using spectro-temporal analysis.

Source: AASIST research + official implementation.

[click here](#)

RawNet2 – End-to-End Anti-Spoofing

Raw waveform-based approach for detecting spoofed/synthetic speech.

Source: ICASSP 2021 / EURECOM.

[click here](#)

Primary Research

47% of Indians exposed to AI voice scams - McAfee

₹1,935 Cr lost to digital-arrest scams in 2025 - MHA / I4C

\$12.5B global contact-center fraud losses in 2024 — Pindrop

SUPPORTING TECHNOLOGIES

Audio Processing

- MFCC
- CQT Spectrogram
- Raw Waveform

Speech & Context

- Whisper → Speech-to-text
- NLP → Urgency/fraud keywords

DATASET & MODEL RESEARCH

ASVspoof 2019 / 2021:
Benchmark dataset for detecting synthetic speech and training models (like RawNet2).

ECAPA-TDNN:
Architecture used for speaker verification and identity comparison.

Features	Our Website	Traditional Solutions	Other Tools (Generic AI)
Consumer Protection	✓	Limited	✗
Hindi + English + Marathi	✓	✗	English Only
"Why Flagged?" Explainability	✓	✗	✗
Voice + Context + NLP Fusion	✓	✗	✗
Timestamp-Level Detection	✓	✗	✗
Telephony + VoIP + IVR Integration	✓	Limited	✗